

EXPERIMENT-10

IMPLEMENTATION OF RSA

AIM

To implement the RSA algorithm for encryption and decryption of a message.

ALGORITHM

1. Choose two prime numbers p and q .
2. Calculate $n = p \times q$.
3. Calculate $\phi(n) = (p-1)(q-1)$.
4. Choose a public key e such that $\gcd(e, \phi(n)) = 1$.
5. Calculate the private key d .
6. Encrypt the message using the public key.
7. Decrypt the ciphertext using the private key.
8. Display the original, encrypted, and decrypted messages.

Encryption: $C = M^e \bmod n$

Decryption: $M = C^d \bmod n$

CODE:

```
main.c
1  #include <stdio.h>
2  int power(int a, int b, int n)
3  {
4      int r = 1;
5      while(b-->0)
6          r = (r * a) % n;
7      return r;
8  }
9  int main()
10 {
11     int p, q, n, phi, e, d, m, c;
12     printf("Enter p and q: ");
13     scanf("%d%d", &p, &q);
14     n = p * q;
15     phi = (p-1) * (q-1);
16     printf("Enter public key e: ");
17     scanf("%d", &e);
18     for(d=1; (d*e)%phi != 1; d++);
19     printf("Enter message: ");
20     scanf("%d", &m);
21     c = power(m, e, n);
22     printf("Public Key : (%d,%d)\n", e, n);
23     printf("Private Key : (%d,%d)\n", d, n);
24     printf("Encrypted : %d\n", c);
25     printf("Decrypted : %d\n", power(c, d, n));
26     return 0;
27 }
```

OUTPUT:

Output

```
Enter p and q: 3 11
Enter public key e: 7
Enter message: 6
Public Key : (7,33)
Private Key : (3,33)
Encrypted : 30
Decrypted : 6

=== Code Execution Successful ===
```

RESULT:

Thus, the RSA algorithm was successfully implemented, and the original message was recovered after encryption and decryption.